Course Review

Lecturer	CLASS CODE	COURSE
Tran Nguyen Lan	**TOAE203**	**Statistics and Probability in Economics and Business**

THE BIG IDEA

The whole course builds one equation

$$Y = \beta_0 + \beta_1 X_1 + \ldots + \varepsilon$$

Y and X

The data sessions prepare the raw material for the equation.

ε

The probability sessions explain why every β comes with error.

β

Tests and regressions are steps of this same equation.

14 sessions in four blocks

■ Data foundations ■ Probability and Sampling ■ Model core ■ Research Process

Lecture 1
Introduction

Lecture 2
Data and Descriptive statistics

Lecture 3
Data graphics

Lecture 4
Probability, Bayes

Lecture 5
Probability models

Lecture 6
Sampling, CLT

Lecture 7
Correlation, simple regression

Lecture 8
Research process, estimation

Lecture 9
Hypothesis testing

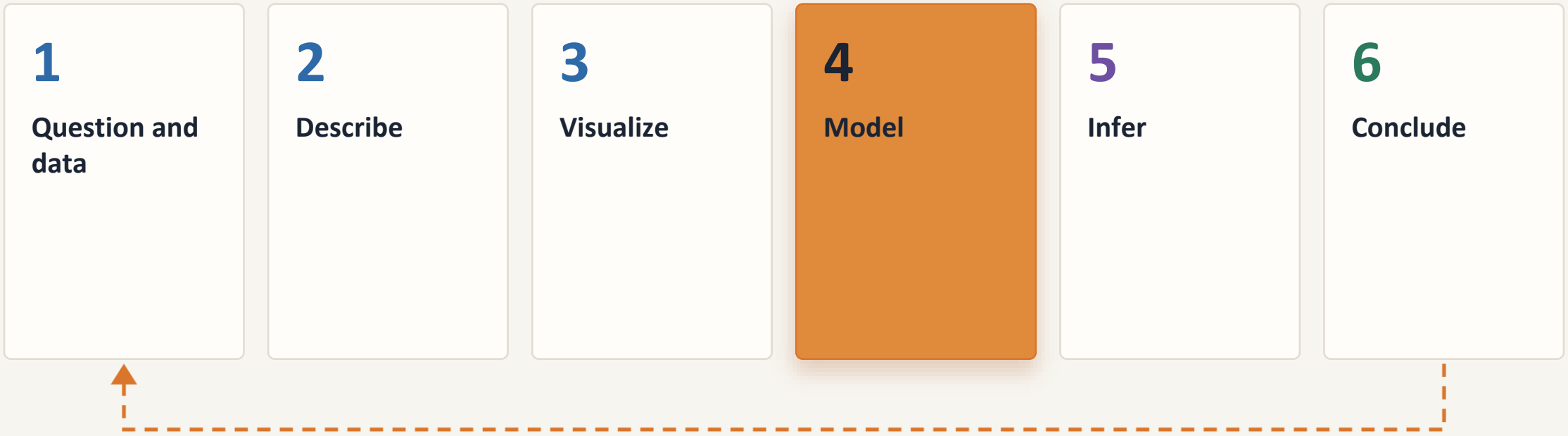
Lecture 10
Two samples, rank tests

Lecture 11
ANOVA → multiple regression

Lecture 12
Special case: Categorical data, χ^2, logarithm, time-series

One question, six steps

Step 4 is the core. Real analysis never runs in one direction.



The loop: a conclusion opens a new question, or sends you back to fix the model

FOUNDATIONS · DATA

Know the type of every variable

CATEGORICAL · NOMINAL

Counts and shares · bar chart · χ^2 test · enters a model as dummies

CATEGORICAL · ORDINAL

Order matters, distances do not · median and ranks · rank-based tests

NUMERICAL · DISCRETE

Countable values · mean and SD · bar chart or histogram

NUMERICAL · CONTINUOUS

Mean, SD, skewness · histogram and scatter · Y or X in a regression

> The type of Y and of each X decides which type of the model applies.

FOUNDATIONS · DATA

Describing one variable: center, spread, shape

Center

Mean uses every value; the median is the middle one. A long tail pulls the mean, not the median.

Spread

Variance, SD, IQR. The variance of Y is SST/(n − 1): the total every model tries to split.

Shape

Skewness and outliers. A right tail (mean above median) suggests modelling ln(Y).

FOUNDATIONS · DATA

Choose the chart by the question

Question	Chart	In the wine data
How is one variable distributed?	Histogram, boxplot	Price and ln(Price)
How do groups compare?	Side-by-side boxplots	Price by region
How do two variables move together?	Scatter plot	Price against Score
How are categories shared?	Bar chart	Wines by variety
How does a value change over time?	Line chart	Not in a cross-section

> Look before you model: the histogram decides log or not; the scatter decides straight or curved.

FOUNDATIONS · PROBABILITY

Probability rules and the conditional trap

COMPLEMENT

$P(\text{not } A) = 1 - P(A)$

ADDITION

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

CONDITIONAL

$P(A \mid B) = P(A \text{ and } B) / P(B)$

INDEPENDENCE

$P(A \text{ and } B) = P(A) \times P(B)$

BAYES

$P(A \mid B) = P(B \mid A) \times P(A) / P(B)$

$P(A \mid B) \neq P(B \mid A)$

A p-value is $P(\text{data this extreme} \mid H_0)$, not the probability that H_0 is true.

Hypothetical screening test: 1% prevalence, 90% sensitivity, 90% specificity. Only about 8% of positive results are true positives.

Regression predicts $E[Y \mid X]$: a mean conditional on X.

FOUNDATIONS · PROBABILITY

Random variables and their distributions

DISCRETE

Binomial(n, p)

Number of successes in n independent trials, each with probability p.

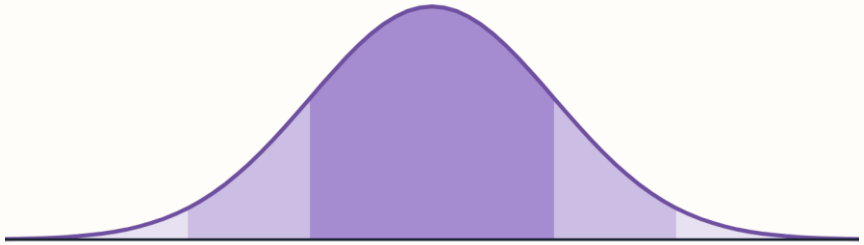
Example: how many of 20 shop visitors buy a bottle.

$E[X] = np$

$Var(X) = np(1 - p)$

CONTINUOUS

Normal(μ, σ) · z = (x − μ) / σ



About 68% of values lie within 1σ of μ, 95% within 2σ, and 99.7% within 3σ.

FOUNDATIONS · PROBABILITY

Sampling distributions and the CLT

Each sample gives a different $\bar{x}$

The distribution of $\bar{x}$ over all possible samples is the sampling distribution. Its standard deviation is the standard error, $SE = \sigma/\sqrt{n}$.

CLT: for large n, $\bar{x}$ is approximately normal, whatever the shape of the population.

Every SE in the course starts here

The Std Error column of the regression output applies the same idea to coefficients.

FOUNDATIONS · INFERENCE

Estimation: estimate ± margin of error

$$\text{estimate} \pm \text{Test Statistics} \times \text{SE}$$

CONFIDENCE INTERVAL

Mean ± Statistics × SE

READING A 95% INTERVAL

Correct: 95% of intervals built this way contain the true mean μ.

Not: 95% of data lie between 2 values of the confidence interval

FOUNDATIONS · INFERENCE · S9

Hypothesis testing: signal over noise

1

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H_0: μ = 25
H_1: μ ≠ 25

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Signal ÷ noise

t = (29.58 − 25) / 1.20 = 3.81

3

p-value

Chance of a t this extreme if H_0 were true: 0.00015

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p < 0.05: reject H_0. The mean price is above $25.

TYPE I ERROR

Rejecting a true H_0. Its probability is α, fixed in fixed in advance.

TYPE II ERROR

Missing a real effect. A larger n makes it less likely.

SIGNIFICANT ≠ IMPORTANT

Read the size of an effect in real units before its p-value.

The model ladder

Each step adds exactly one thing. The separate tests are one equation with a longer right-hand side.

Y	X	Equation	Familiar names	What's new
Continuous	No X	$Y = \beta_0 + \varepsilon$	Mean, CI, one-sample t	A mean has error
Continuous	One dummy X	$Y = \beta_0 + \beta_1 D + \varepsilon$	Two-sample t	Comparison
Continuous	One X, 3+ groups	$Y = \beta_0 + \Sigma\, \beta_j D_j + \varepsilon$	One-way ANOVA	SST split, F
Continuous	One continuous X	$Y = \beta_0 + \beta_1 X + \varepsilon$	Correlation, simple regression	Slope
Continuous	Several continuous X	$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \varepsilon$	Multiple regression	Control
Categorical	Categorical X	$P(Y = 1 \mid X)$	Crosstab, Pearson's χ^2	Counts, not means

CONTINUOUS Y

- **No X**
- One dummy X
- One X, 3+ groups
- One continuous X
- Several continuous X

CATEGORICAL Y

- Categorical X

CONTINUOUS Y · NO X · L2 · L6 · L8 · L9

Even one number has error

$$Y = \beta_0 + \varepsilon$$

COEFFICIENT MEANS

$\beta_0 = \bar{y}$, the sample mean

FAMILIAR NAMES

Descriptive statistics, CI for μ, one-sample t

> **What's new:** even a mean changes from sample to sample, so it has a standard error and a confidence interval.

CONTINUOUS Y

- No X
- **One dummy X**
- **One X, 3+ groups**
- One continuous X
- Several continuous X

CATEGORICAL Y

- Categorical X

CONTINUOUS Y · ONE CATEGORICAL X · L10 · L12

Comparing groups is regression on dummies

ONE DUMMY X (2 GROUPS)

$$Y = \beta_0 + \beta_1 D + \varepsilon$$

β_1 = difference in means between the two groups

Two-sample t · Welch · Mann–Whitney

ONE X WITH 3+ GROUPS

$$Y = \beta_0 + \Sigma\, \beta_j D_j + \varepsilon$$

β_j = difference from the reference group

One-way ANOVA · Tukey · Kruskal–Wallis

What's new: comparison. With 3+ groups we also split SST = SSR + SSE and use the F statistic.

CONTINUOUS Y

- No X
- One dummy X
- One X, 3+ groups
- **One continuous X**
- **Several continuous X**

CATEGORICAL Y

- Categorical X

CONTINUOUS Y · CONTINUOUS X · L7 · L12

From a slope to holding others fixed

ONE CONTINUOUS X

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

β_1 = change in Y when X rises by one unit

Pearson correlation · simple regression

SEVERAL CONTINUOUS X

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \varepsilon$$

β_j = effect of X_j holding the other X constant

Multiple regression · F for groups of variables

> **What's new:** a slope instead of a group difference, then control. Leave out a confounder and the coefficient is biased.

CONTINUOUS Y

- No X
- One dummy X
- One X, 3+ groups
- One continuous X
- Several continuous X

CATEGORICAL Y

- **Categorical X**

CATEGORICAL Y · CATEGORICAL X · L4 · L11

When Y is a count: Pearson's χ^2

$$\chi^2 = \Sigma\,(O - E)^2 / E$$

THE QUESTION

Are the two categorical variables independent?

FROM THE CROSSTAB

Compare observed counts O with counts E expected under independence

> **What's new:** counts instead of means. Same role as F: a large signal relative to random noise rejects H_o.

The regression output holds the course

REGRESSION STATISTICS

Cell	Symbol	Session
Multiple R	|r|	L7
R Square	SSR/SST	L7 · L11
Adjusted R^2	$\bar{R}^2$	L11
Standard Error	se	L7
Observations	n	L6

ANOVA

	ÑN	SS	MS	F	Significance F
Regression	k	SSR	MSR	MSR/MSE	p
Residual	n−k−1	SSE	MSE		
Total	n−1	SST			
Session	L11	L11	L11	L11	L9

COEFFICIENTS

	FÖÑNB	Std Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	b_0	SE(b_0)	b_0/SE	p	$b_0 - t \cdot SE$	$b_0 + t \cdot SE$
X_1	b_1	SE(b_1)	b_1/SE	p	$b_1 - t \cdot SE$	$b_1 + t \cdot SE$
Session	L7 · L11	L6	L9	L9	L8	L8

Four keys to reading the output

STD ERROR ← L6

Standard error

Same idea as σ/√n: a coefficient computed from a sample sample moves when the sample changes.

LOWER / UPPER 95% ← L8

Confidence interval

Still estimate ± t·SE, with the coefficient as the estimate. estimate.

T STAT, P-VALUE ← L9

Testing H_0: β = 0

The numerator is the signal, the denominator is the noise.

ANOVA BLOCK ← L11

SST = SSR + SSE

With only dummies as X, this is exactly the one-way ANOVA table.

LINKS UP

What each block feeds into the next

L2 → no-X model

The sample mean is β_0 of the model with no X. The variance of Y is the SST that every later model tries to split up.

L3 → step 4 (model)

Histograms before and after logs decide whether Y enters as is or in logs. A scatter shows whether the relationship is straight.

L4 → steps 4 and 5

Regression is a conditional mean E[Y | X]. A p-value is a probability conditional on H_0. Flipping the condition is the same mistake in both.

L5–6 → step 5 (infer)

The CLT is why b_1 is close to normal for large n, even if Y is skewed. SE shrinks with √n, so large n makes tiny effects "significant".

LINKS UP

What each block feeds into the next

L7 → several-X model

Confounders and Simpson's paradox are the problem. Multiple regression in L12 is a partial fix: put the confounder into the equation.

L10 → dummy X, 3+ groups

The decision tree for choosing a test is really choosing a model: by type of X, number of X, and whether assumptions hold.

L12 → categorical Y

χ^2 is the counterpart of F when Y is a count: it also weighs a signal (O − E) against random variation.

L9 → step 6 (conclude)

“Significant” is not “important”. Read a coefficient in real units before talking about its p-value.

L12 · BEYOND THE CORE

Three ways past the basic equation

L12

Change the form

Y or X enters the equation in logs logs instead of raw units.

L12

Change the type of Y

Binary Y: model P(Y = 1 | X) with with logistic regression.

L12

Drop independence

Time series: today's observation depends on yesterday's.

> All three keep the frame Y = f(X) + ε. This is the door to the courses that come next.